

```
-- =====
```

```
-- 1. User Upcoming Events
```

```
-- =====
```

```
SELECT
    u.full_name,
    e.title,
    e.city,
    e.start_date
FROM Users u
JOIN Registrations r
ON u.user_id = r.user_id
JOIN Events e
ON r.event_id = e.event_id
WHERE e.status = 'upcoming'
AND u.city = e.city
ORDER BY e.start_date;
```

```
-- =====
```

```
-- 2. Top Rated Events
```

```
-- =====
```

```
SELECT
    e.title,
    AVG(f.rating) AS average_rating,
    COUNT(f.feedback_id) AS total_feedbacks
FROM Events e
```

```
JOIN Feedback f
ON e.event_id = f.event_id
GROUP BY e.event_id, e.title
HAVING COUNT(f.feedback_id) >= 10
ORDER BY average_rating DESC;
```

```
-- =====
```

```
-- 3. Inactive Users
```

```
-- =====
```

```
SELECT *
FROM Users
WHERE user_id NOT IN
(
    SELECT DISTINCT user_id
    FROM Registrations
    WHERE registration_date >= CURDATE() - INTERVAL 90 DAY
);
```

```
-- =====
```

```
-- 4. Peak Session Hours
```

```
-- =====
```

```
SELECT
    e.title,
    COUNT(s.session_id) AS total_sessions
```

```
FROM Events e
JOIN Sessions s
ON e.event_id = s.event_id
WHERE TIME(s.start_time) BETWEEN '10:00:00' AND '12:00:00'
GROUP BY e.event_id, e.title;
```

```
-- =====
```

```
-- 5. Most Active Cities
```

```
-- =====
```

```
SELECT
    u.city,
    COUNT(DISTINCT r.user_id) AS total_users
FROM Users u
JOIN Registrations r
ON u.user_id = r.user_id
GROUP BY u.city
ORDER BY total_users DESC
LIMIT 5;
```

```
-- =====
```

```
-- 6. Event Resource Summary
```

```
-- =====
```

```
SELECT
    e.title,
```

```
        COUNT(r.resource_id) AS total_resources
FROM Events e
LEFT JOIN Resources r
ON e.event_id = r.event_id
GROUP BY e.event_id, e.title;
```

```
-- =====
```

```
-- 7. Low Feedback Alerts
```

```
-- =====
```

```
SELECT
    u.full_name,
    e.title,
    f.rating,
    f.comments
FROM Feedback f
JOIN Users u
ON f.user_id = u.user_id
JOIN Events e
ON f.event_id = e.event_id
WHERE f.rating < 3;
```

```
-- =====
```

```
-- 8. Sessions per Upcoming Event
```

```
-- =====
```

```
SELECT
    e.title,
    COUNT(s.session_id) AS total_sessions
FROM Events e
LEFT JOIN Sessions s
ON e.event_id = s.event_id
WHERE e.status = 'upcoming'
GROUP BY e.event_id, e.title;
```

```
-- =====
-- 9. Organizer Event Summary
-- =====
```

```
SELECT
    u.full_name,
    e.status,
    COUNT(e.event_id) AS total_events
FROM Users u
JOIN Events e
ON u.user_id = e.organizer_id
GROUP BY u.full_name, e.status;
```

```
-- =====
-- 10. Feedback Gap
-- =====
```

```
SELECT DISTINCT
    e.title
FROM Events e
JOIN Registrations r
ON e.event_id = r.event_id
LEFT JOIN Feedback f
ON e.event_id = f.event_id
WHERE f.feedback_id IS NULL;
```

```
-- =====
-- 11. Daily New User Count
-- =====
```

```
SELECT
    registration_date,
    COUNT(user_id) AS total_users
FROM Users
WHERE registration_date >= CURDATE() - INTERVAL 7 DAY
GROUP BY registration_date;
```

```
-- =====
-- 12. Event with Maximum Sessions
-- =====
```

```
SELECT
    e.title,
```

```

COUNT(s.session_id) AS total_sessions
FROM Events e
JOIN Sessions s
ON e.event_id = s.event_id
GROUP BY e.event_id, e.title
HAVING COUNT(s.session_id) =
(
SELECT MAX(session_count)
FROM
(
SELECT COUNT(session_id) AS session_count
FROM Sessions
GROUP BY event_id
) AS temp
);

```

```

-- =====
-- 13. Average Rating per City
-- =====

```

```

SELECT
    e.city,
    AVG(f.rating) AS average_rating
FROM Events e
JOIN Feedback f
ON e.event_id = f.event_id
GROUP BY e.city;

```

```
-- =====
```

```
-- 14. Most Registered Events
```

```
-- =====
```

```
SELECT
    e.title,
    COUNT(r.registration_id) AS total_registrations
FROM Events e
JOIN Registrations r
ON e.event_id = r.event_id
GROUP BY e.event_id, e.title
ORDER BY total_registrations DESC
LIMIT 3;
```

```
-- =====
```

```
-- 15. Event Session Time Conflict
```

```
-- =====
```

```
SELECT
    s1.event_id,
    s1.title AS session1,
    s2.title AS session2
FROM Sessions s1
JOIN Sessions s2
ON s1.event_id = s2.event_id
```

AND s1.session_id < s2.session_id

AND s1.start_time < s2.end_time

AND s1.end_time > s2.start_time;

-- =====

-- 16. Unregistered Active Users

-- =====

SELECT *

FROM Users

WHERE registration_date >= CURDATE() - INTERVAL 30 DAY

AND user_id NOT IN

(

SELECT user_id

FROM Registrations

);

-- =====

-- 17. Multi-Session Speakers

-- =====

SELECT

speaker_name,

COUNT(session_id) AS total_sessions

FROM Sessions

GROUP BY speaker_name

```
HAVING COUNT(session_id) > 1;
```

```
-- =====
```

```
-- 18. Resource Availability Check
```

```
-- =====
```

```
SELECT
```

```
    e.title
```

```
FROM Events e
```

```
LEFT JOIN Resources r
```

```
ON e.event_id = r.event_id
```

```
WHERE r.resource_id IS NULL;
```

```
-- =====
```

```
-- 19. Completed Events with Feedback Summary
```

```
-- =====
```

```
SELECT
```

```
    e.title,
```

```
    COUNT(DISTINCT r.registration_id) AS total_registrations,
```

```
    AVG(f.rating) AS average_rating
```

```
FROM Events e
```

```
LEFT JOIN Registrations r
```

```
ON e.event_id = r.event_id
```

```
LEFT JOIN Feedback f
```

```
ON e.event_id = f.event_id
```

```
WHERE e.status = 'completed'
```

```
GROUP BY e.event_id, e.title;
```

```
-- =====
```

```
-- 20. User Engagement Index
```

```
-- =====
```

```
SELECT
```

```
    u.full_name,
```

```
    COUNT(DISTINCT r.event_id) AS attended_events,
```

```
    COUNT(DISTINCT f.feedback_id) AS feedbacks_given
```

```
FROM Users u
```

```
LEFT JOIN Registrations r
```

```
ON u.user_id = r.user_id
```

```
LEFT JOIN Feedback f
```

```
ON u.user_id = f.user_id
```

```
GROUP BY u.user_id, u.full_name;
```

```
-- =====
```

```
-- 21. Top Feedback Providers
```

```
-- =====
```

```
SELECT
```

```
    u.full_name,
```

```
    COUNT(f.feedback_id) AS total_feedbacks
```

```
FROM Users u
```

```
JOIN Feedback f
ON u.user_id = f.user_id
GROUP BY u.user_id, u.full_name
ORDER BY total_feedbacks DESC
LIMIT 5;
```

```
-- =====
```

```
-- 22. Duplicate Registrations Check
```

```
-- =====
```

```
SELECT
    user_id,
    event_id,
    COUNT(*) AS duplicate_count
FROM Registrations
GROUP BY user_id, event_id
HAVING COUNT(*) > 1;
```

```
-- =====
```

```
-- 23. Registration Trends
```

```
-- =====
```

```
SELECT
    YEAR(registration_date) AS year,
    MONTH(registration_date) AS month,
    COUNT(registration_id) AS total_registrations
```

```
FROM Registrations

WHERE registration_date >= CURDATE() - INTERVAL 12 MONTH

GROUP BY YEAR(registration_date), MONTH(registration_date)

ORDER BY year, month;
```

```
-- =====
```

```
-- 24. Average Session Duration per Event
```

```
-- =====
```

```
SELECT

    e.title,

    AVG(

        TIMESTAMPDIFF(

            MINUTE,

            s.start_time,

            s.end_time

        )

    ) AS average_duration_minutes

FROM Events e

JOIN Sessions s

ON e.event_id = s.event_id

GROUP BY e.event_id, e.title;
```

```
-- =====
```

```
-- 25. Events Without Sessions
```

```
-- =====
```

```
SELECT
    e.title
FROM Events e
LEFT JOIN Sessions s
ON e.event_id = s.event_id
WHERE s.session_id IS NULL;
```